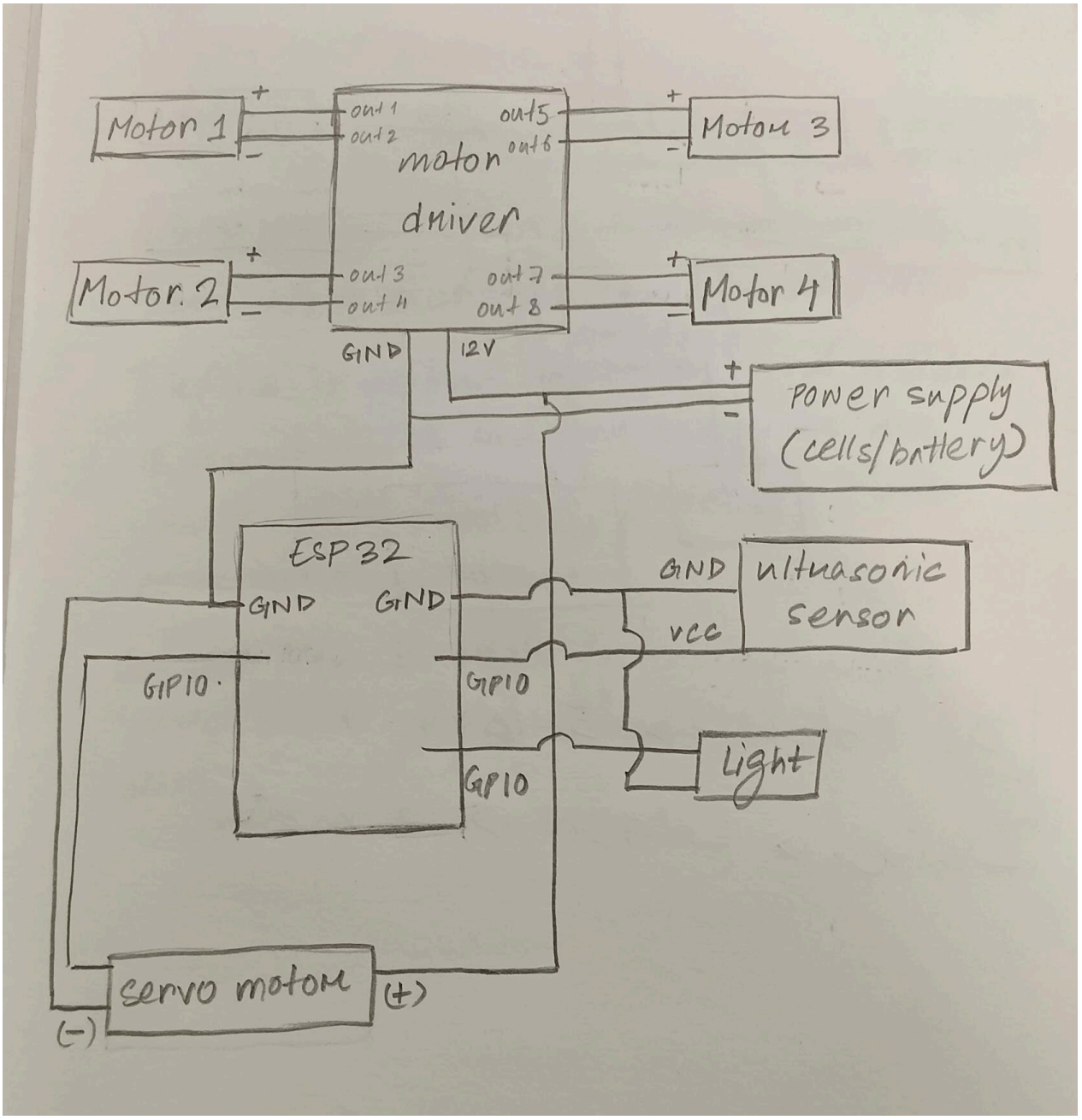
ODT Summative 2

Idea Brief :

The proposed concept is a robotic delivery vehicle designed to autonomously transport small parcels along predefined routes. The system consists of a four-wheeled robotic chassis equipped with a mounted storage compartment for carrying delivery items. Each wheel is driven by an individual DC motor, which is controlled through a motor driver interfaced with an ESP32 microcontroller. The robot is powered by a battery system providing a total output of 24 volts. For navigation and safety, the robot utilizes ultrasonic sensors to detect and avoid obstacles in its path.

The ESP32 is programmed with a set of predetermined pathways, enabling structured and reliable movement between delivery points. A mobile application, connected to the ESP32 via Bluetooth, allows users to select from a catalogue of available routes. Once a path is selected, the robot autonomously follows the chosen route while actively avoiding obstacles. In operation, a user at the starting point can place a parcel in the storage compartment, select the desired destination path through the application, and initiate delivery, allowing therobot to transport the parcel efficiently to the specified endpoint.

Basic Idea Sketch



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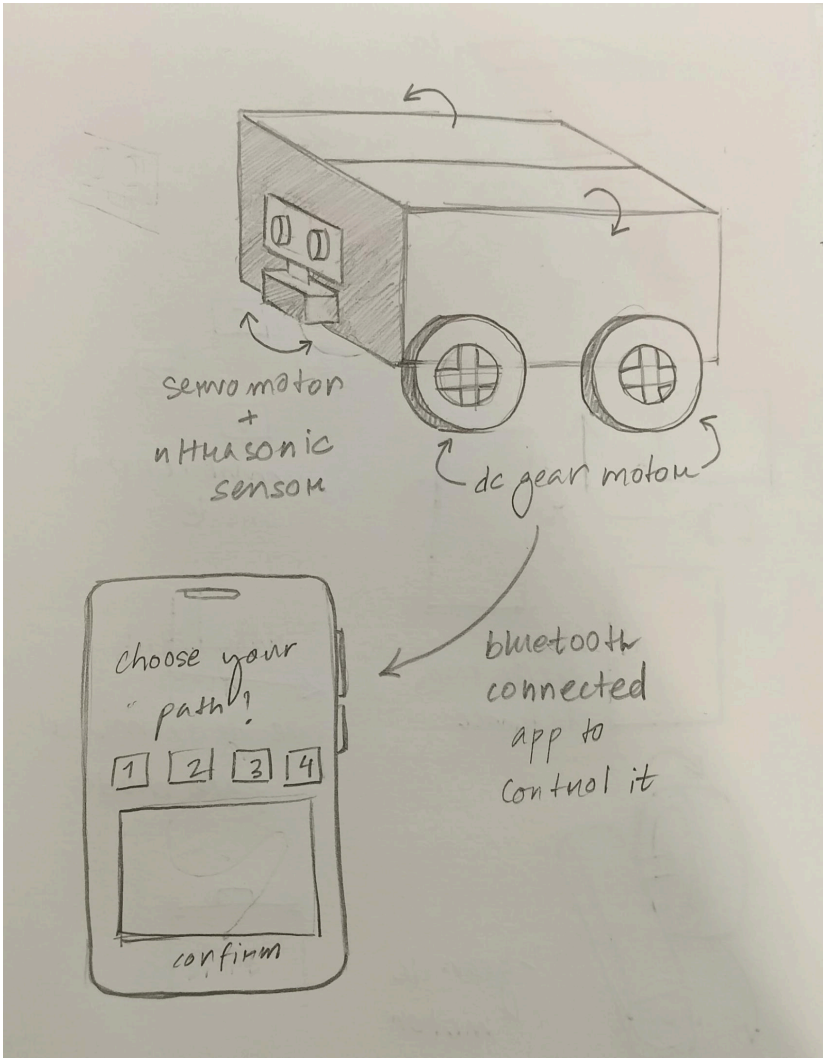
servo motor
+
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sensor
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connected
app to
control it
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path!
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